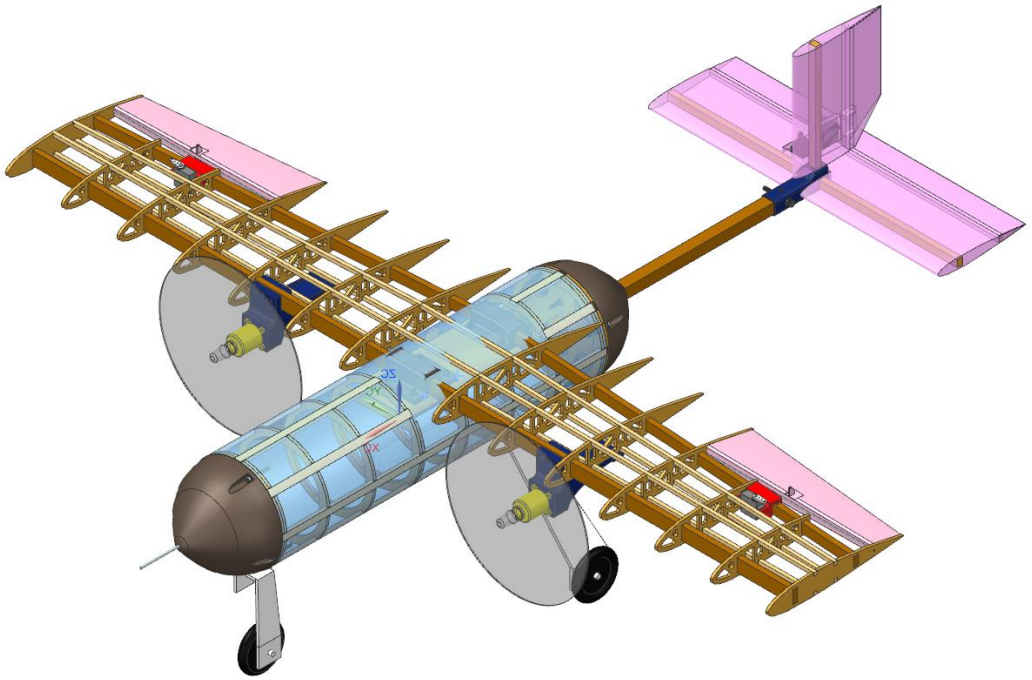
Senior Capstone: Aircraft Design



Problem: Design, build, and test RC aircraft capable of carrying payload and recording critical flight data

Method:

- Design from mission requirements, MATLAB based math models, and Siemens NX modeling
- Manufacture with laser cut plywood, custom 3D printed mounts and brackets
- Transfer aerodynamic loads across airframe via secure attachments, bolts, fasteners

Solution:

- Functional aircraft capable of stable flight without feedback control (full manual piloting)
- Successful data collection allowing for analysis of anomaly during flight and comparison to predicted performance

CFD Analysis: Supersonic Inlet

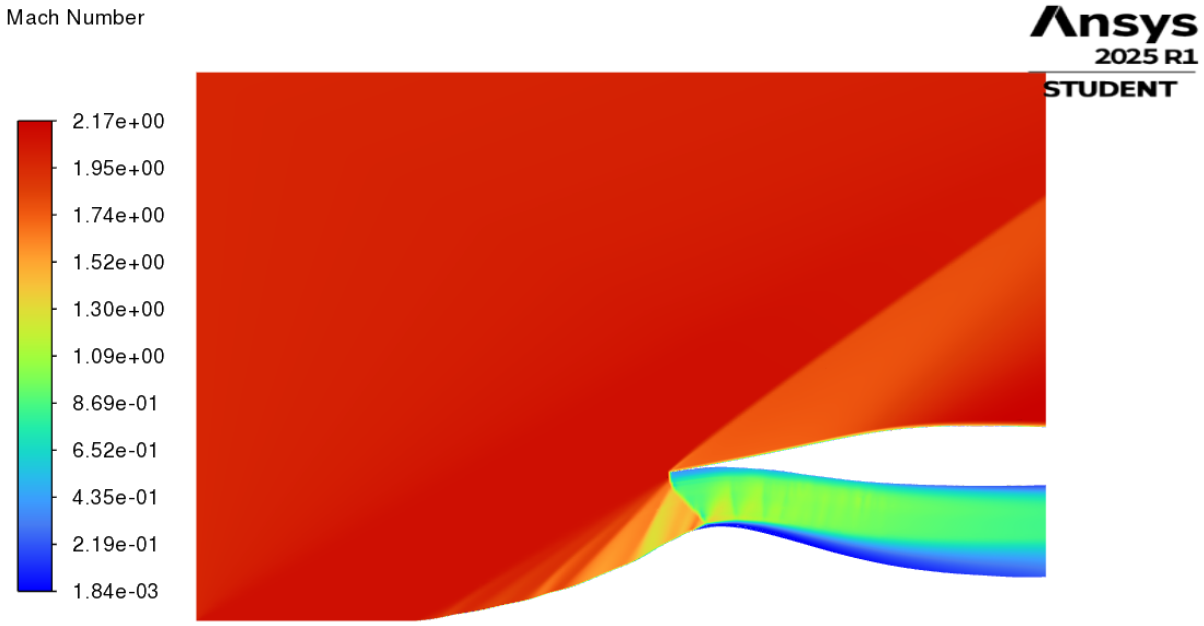
Goal: Perform CFD analysis on supersonic inlet under different flow cases

Method:

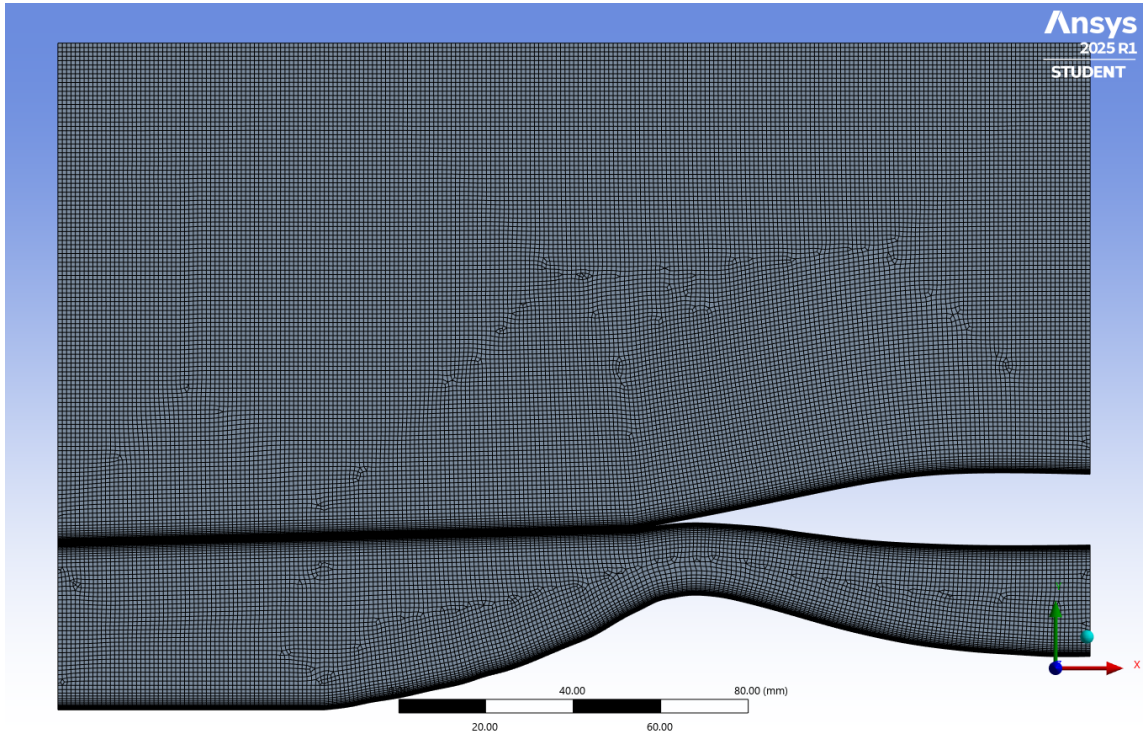
- Define reference flow and obtain solutions on coarse grid with appropriate y1/y+ value
- Refine grid using refinement factor to perform grid study
- Solve using density-based solver with second order methods and AGR to resolve shock locations

Result:

- Estimated inlet performance in off-design conditions (non-zero angle of attack, leeward side)
- Grid refinement study demonstrating tradeoff between solution accuracy and computational cost



mach-contours



Firefighting RC Aircraft

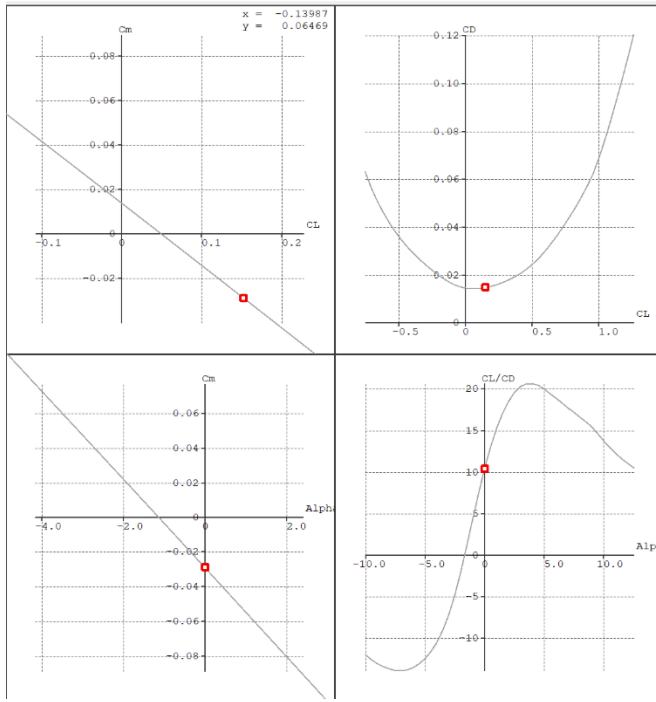
Problem: Create a prototype that supports the integration of electronics that collects data on surroundings

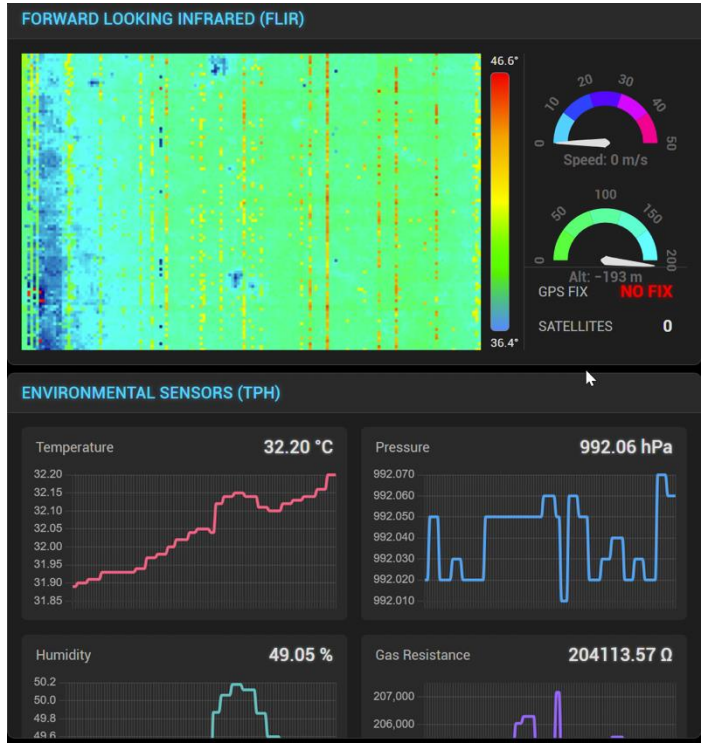
Method:

- Design from first principles and mission requirements, revise based on initial flight test data and stability simulations
- Build using lightweight and durable material
- Solder connectors and wires to support electronics

Solution:

- Working prototype capable of smooth flying
- Successful integration of TPH sensor and thermal camera for live telemetry and data collection





FEA: Integrated Truss Structure

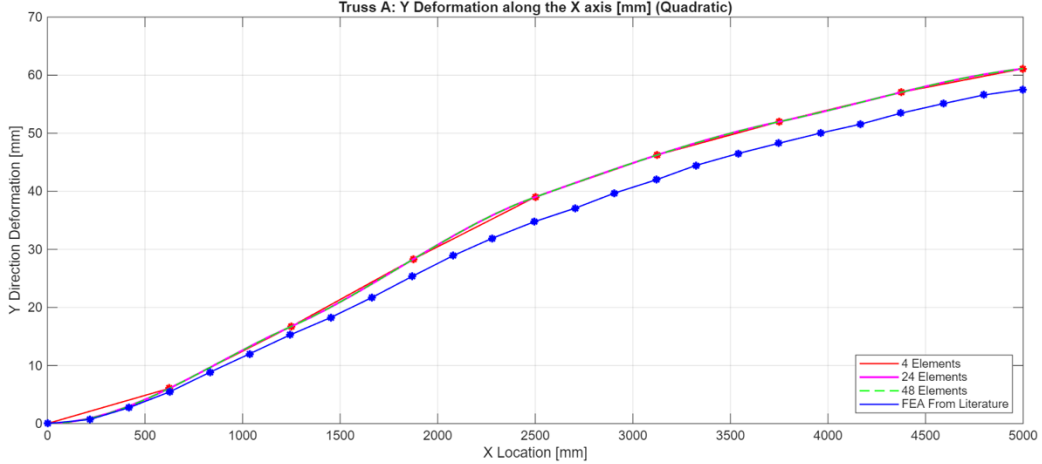
Goal: Predict deformation of truss structure under different load scenarios

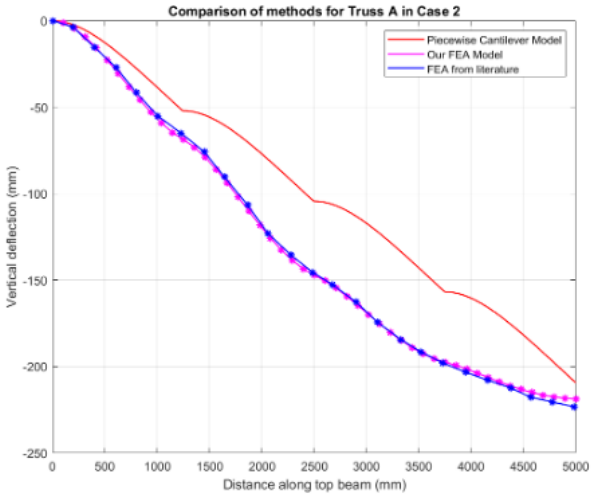
Method:

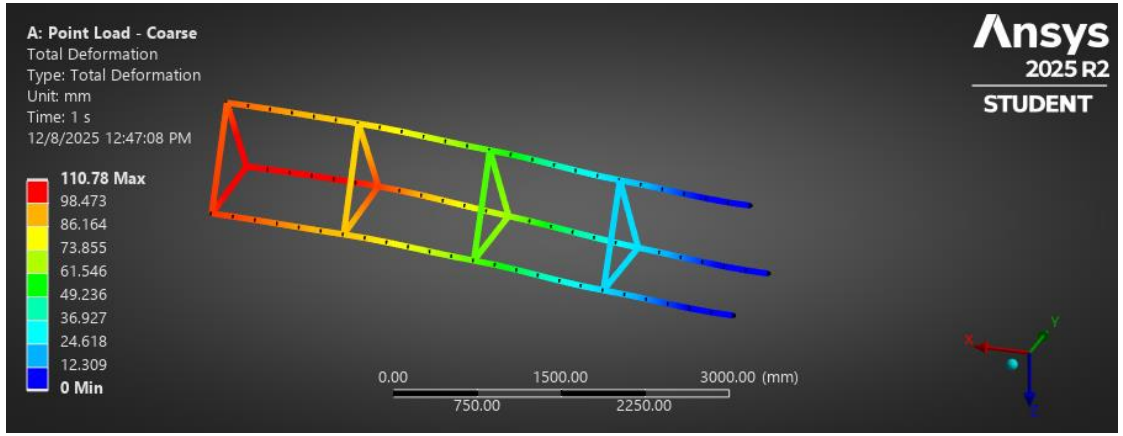
- Derive first-order approximation from Euler-Bernoulli beam theory
- Develop finite-element model using ANSYS FEA suite
- Obtain solutions for each load scenario

Result:

- Comparison of simple approximation with full degree of freedom finite-element model and solution from literature
- Accurate prediction of deformation under load







Conceptual Aircraft Design

Problem: Produce a conceptual design to meet customer requirements

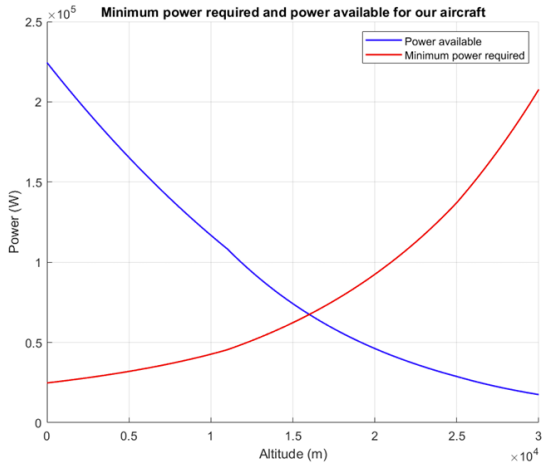
Method:

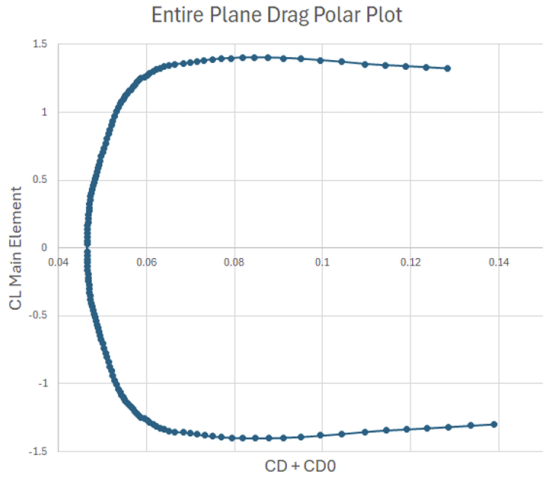
- Define mission profile and requirements
- Choose configuration based on historical data and weighted decision matrices
- Iteratively size the aircraft using MATLAB scripts

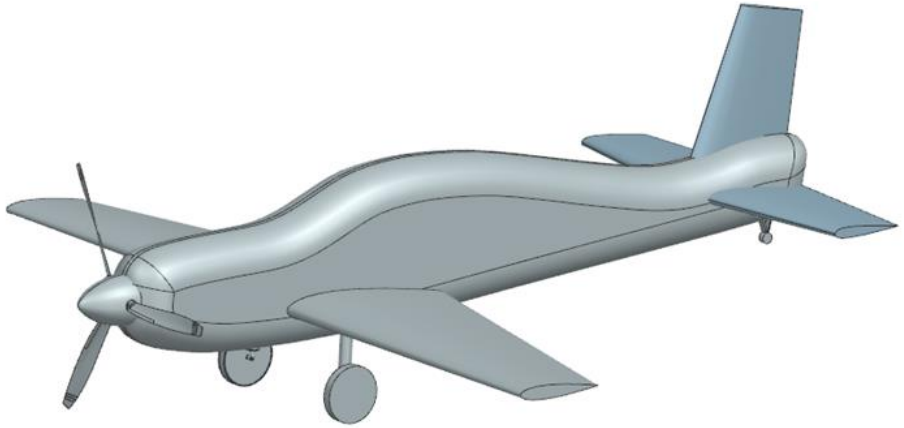
Solution:

- Converged, conceptual aircraft configuration optimized for the mission
- Comprehensive technical report detailing overall design process and outlining next steps in prototyping and testing

Parameter	Symbol	Value
Wingspan	b	6.12 m
Wing area	S	6.192 m^2
Wing aspect ratio	AR_w	6.05
Tip chord	c_{tip}	0.675 m
Root chord	c_{root}	1.35 m
MAC	c_{bar}	1.05 m
Winglets?	N/A	No
Landing gear type?	N/A	Fixed
Prop. Efficiency	η_p	0.86
Empty weight	m_{empty}	517.5 kg
MTOW	m_{TO}	683 kg
Length	l	5.17 m
Height	h	1.97 m
V-Stab area	S_{VT}	0.816 m^2
H-Stab area	S_{HT}	1.5 m^2





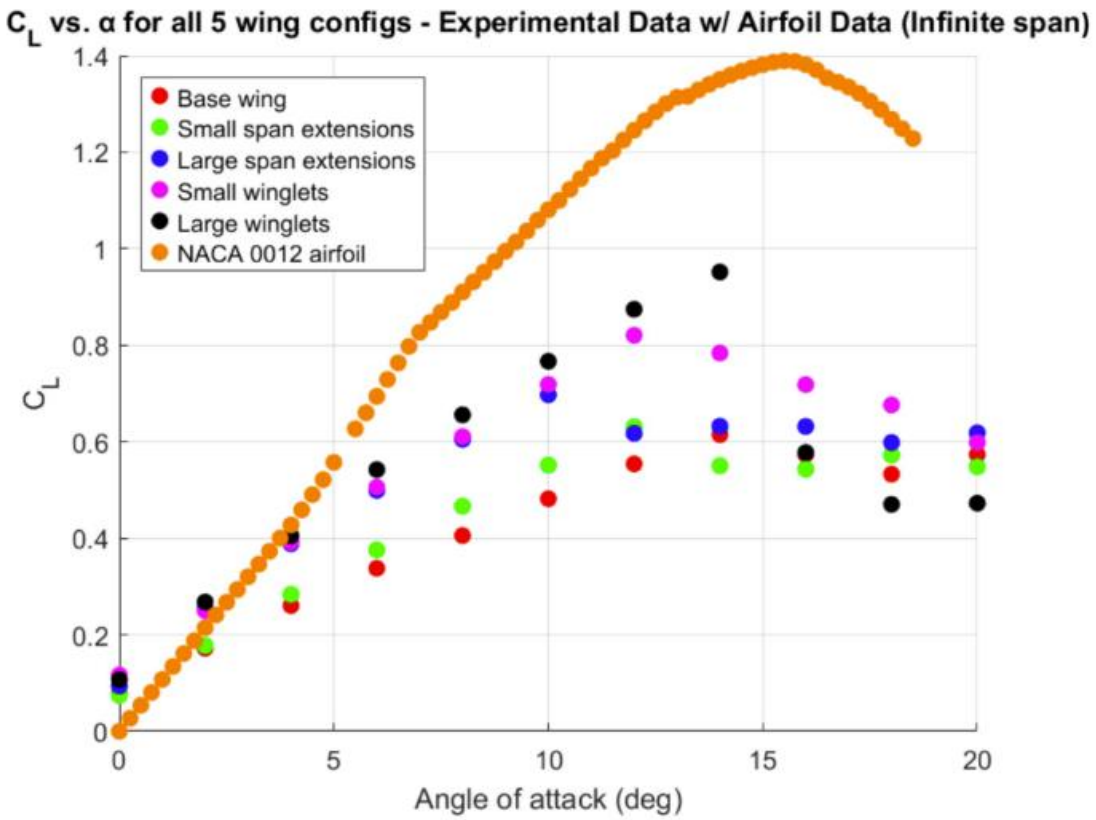


Finite Wing Wind Tunnel Test

Problem: Compare aerodynamic data of wing to that of its airfoil

Method:

- Select an airfoil and create model of wing
- Configure and operate low-speed wind tunnel for testing
- Collect data of various wing spans and winglet configurations using load cells



Results:

- Visualized the effect of wing-tip vortices on lift
- Demonstrated how winglets can improve aerodynamic performance
- Increased understanding of how error can impact experimental data

Rocket Fin Analysis

Problem: Analyze aerodynamic data of varying fin geometry on model rocket in low-speed environment (“launch”)

Method:

- Generate fin geometries using CAD software
- Configure and operate low-speed wind tunnel for testing
- Collect data using load cells

Results:

- Obtained meaningful drag data
- Presented experimental results in comprehensive lab report

Figure 1.1: Model rocket without fins

Figure 1.2: Model rocket with small fins

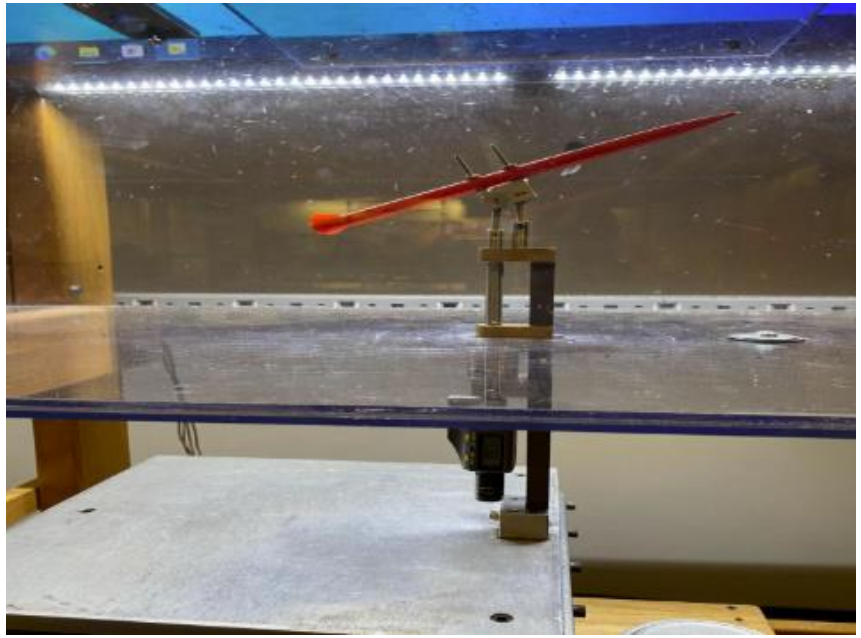



Figure 1.3: Model rocket with large fins